Solid State and Devices

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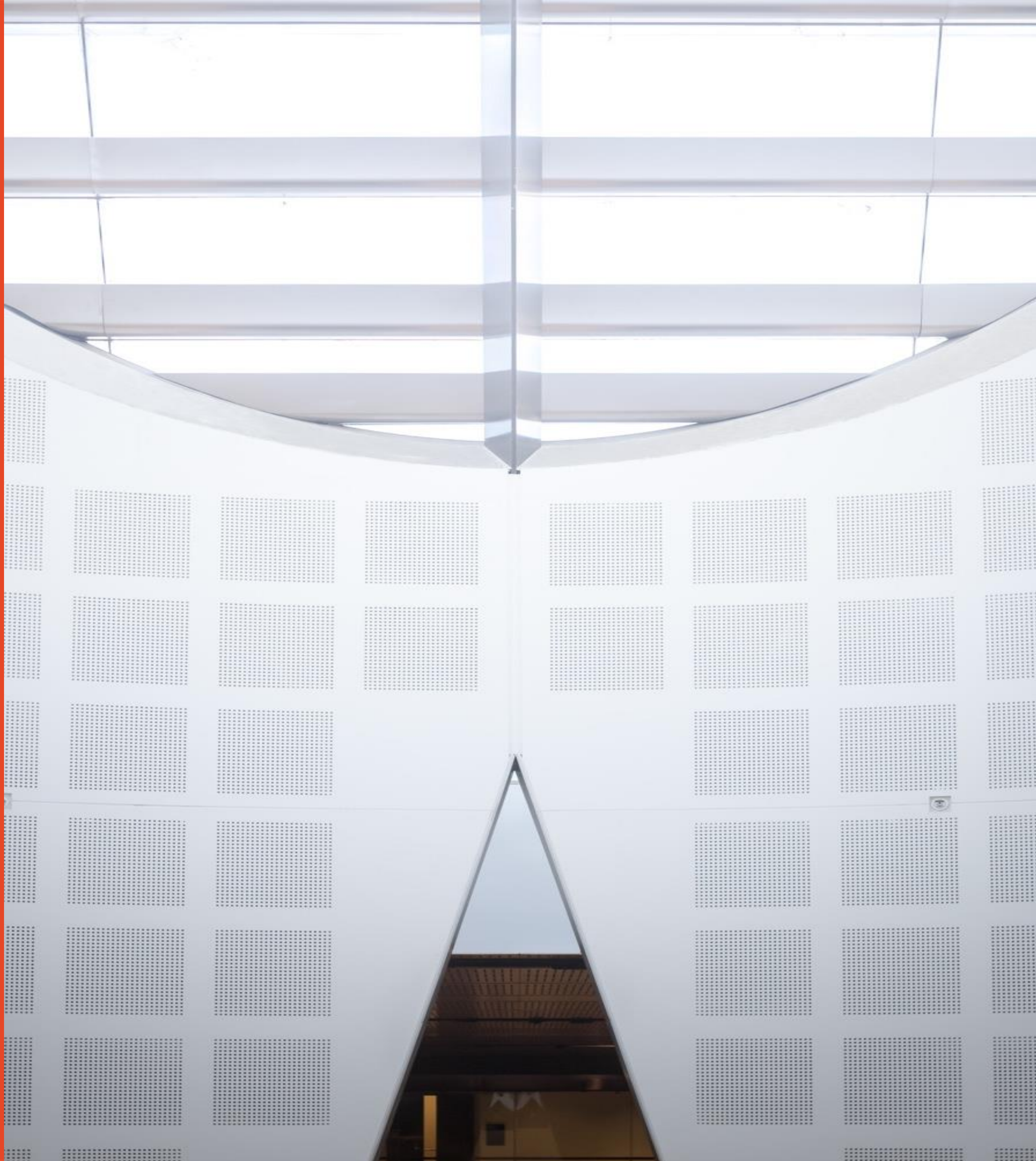
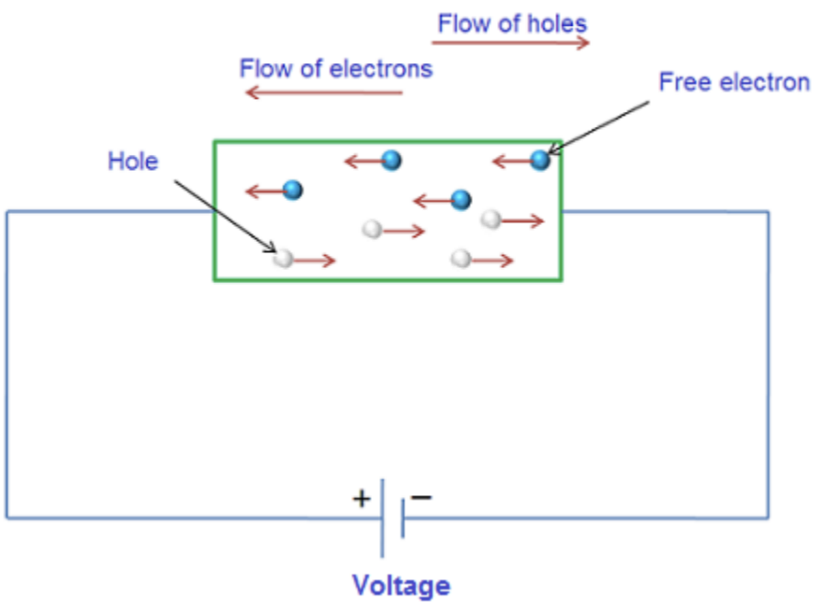


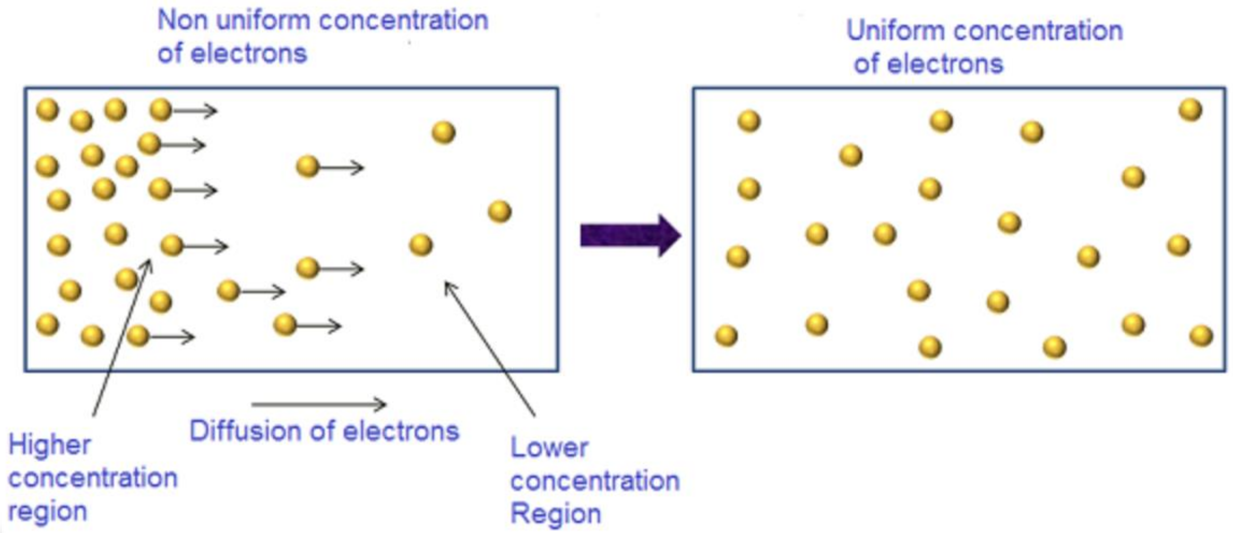
Carrier Transport

"Drift"
The movement of carrier due to electric field (E)



Force Driven

"Diffusion"
The flow of carrier due to density gradients (dn/dx)



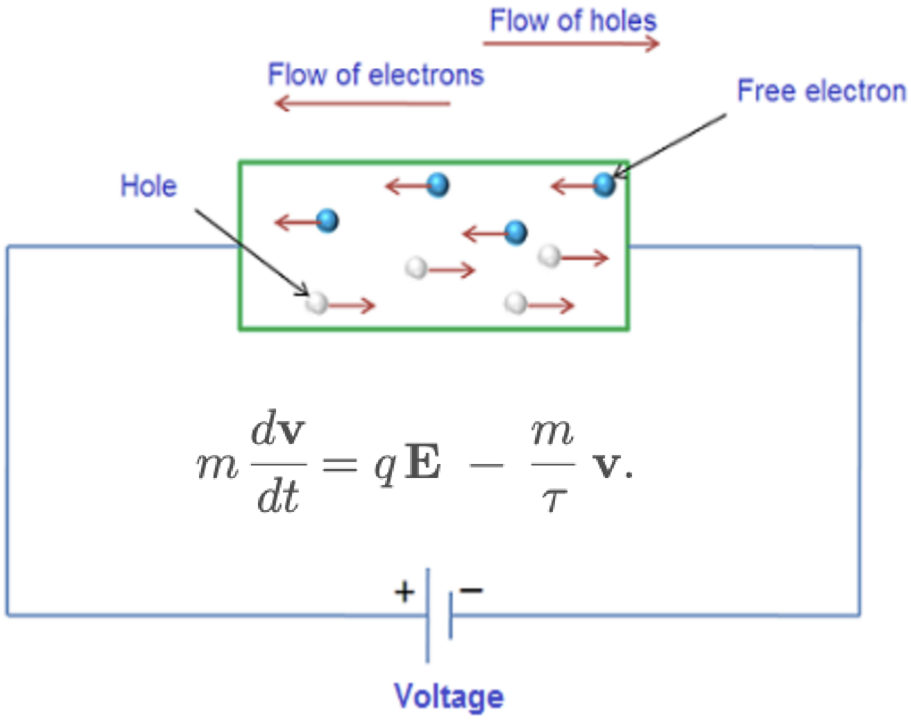
Entropic process

Drift Current

Carrier Motion under an Electric Field

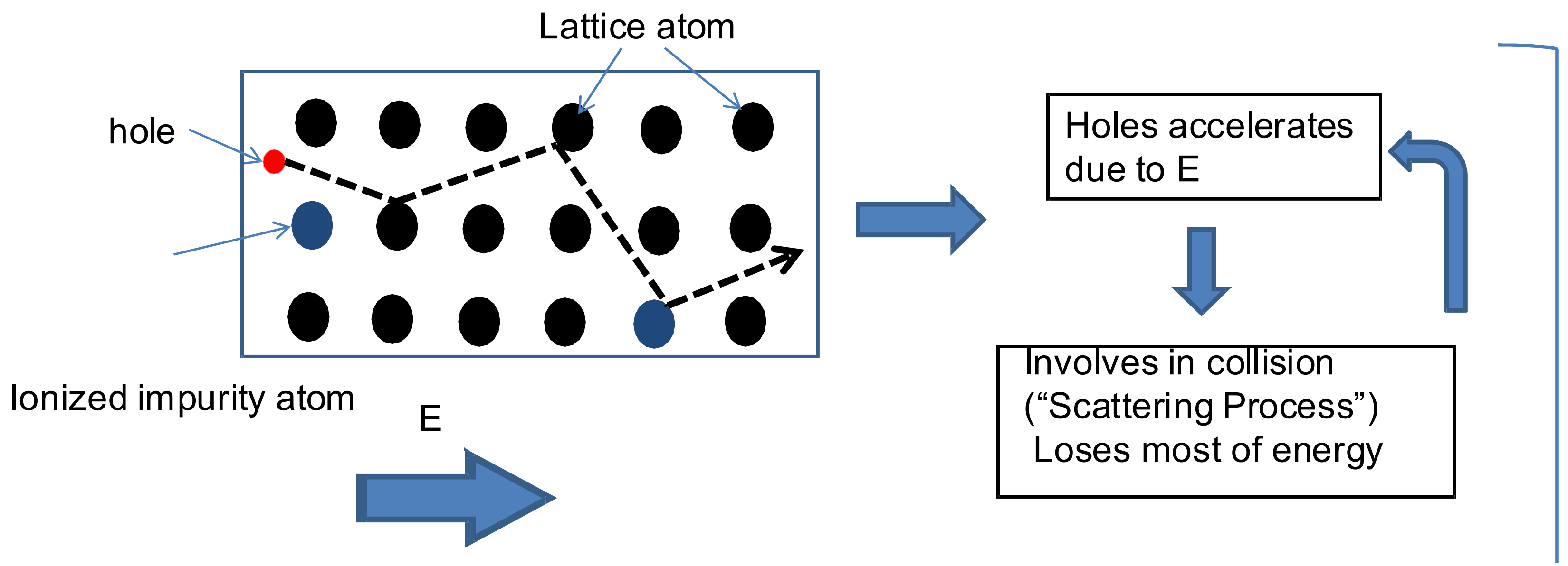
Core Concept
Drift is the net motion of carriers (electrons, holes) in response to an applied electric field **E**.

- **Force on Carrier**: $\mathbf{F} = q\mathbf{E}$, where $q = -e$ for electrons, $q = +e$ for holes; e is positive elementary charge.
- **Equation of Motion**: $m_c^* \frac{d\mathbf{v}_d}{dt} = q\mathbf{E}$ (Newton's second law, ignoring scattering initially).
- **Scattering Effects**: Collisions with lattice phonons and impurities randomize carrier momentum.



$$m\frac{d\mathbf{v}}{dt} = q\,\mathbf{E} - \frac{m}{\tau}\,\mathbf{v}.$$

Acceleration is counterbalanced by collisions during motion (term proportional to velocity)



$$m\frac{d\mathbf{v}}{dt} = q\mathbf{E} - \frac{m}{\tau}\mathbf{v}.$$

the solution is

$$v(t) = v_{ss} + [v(0) - v_{ss}]\, e^{-t/\tau},$$

$$\boxed{v_{ss} = \frac{q\tau}{m} E \equiv \mu E.}$$

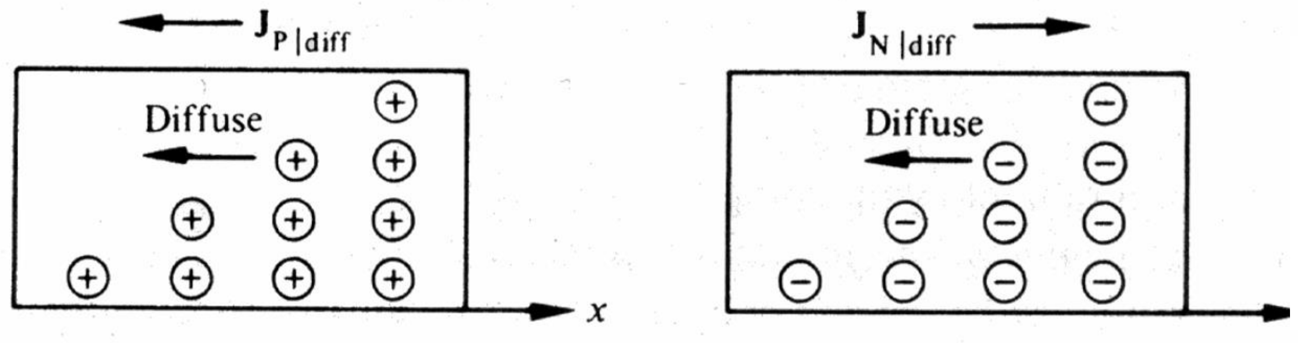
Gain **average drift velocity**, v_{dp}

$$\boxed{\mathbf{v}_d = \pm\mu\mathbf{E}}$$

μ; mobility (unit; cm^2/Vs)
Describes how well a carrier move due to E

Drift Current

Drift velocity is defined as the net velocity of free electrons moving randomly in a conductor due to an electric field. These electrons move at different speeds and directions. When an electric field is applied, they experience a force that aligns them towards the field direction.



$$J_n = -en\vec{v}_{dn} = en\mu_n \vec{E}, \quad J_p = ep\vec{v}_{dp} = ep\mu_p \vec{E}$$

$$\boxed{J_{\text{drift}} = e(n\mu_n + p\mu_p)\vec{E}}$$

At Steady State

$$v_d = \left(\frac{dv}{dt}\right)\tau = -\frac{eE}{m^*}\tau$$

$$\boxed{\mu = \frac{v_d}{E} = \frac{e\tau}{m^*}}$$

$$\boxed{\tau(T, E, n)}$$

Function of E field, Temperature and carrier concentration

•Longer τ ⇒ carriers move farther before scattering ⇒ **higher mobility**
•Shorter τ ⇒ more frequent scattering ⇒ **lower mobility**

Relaxation time

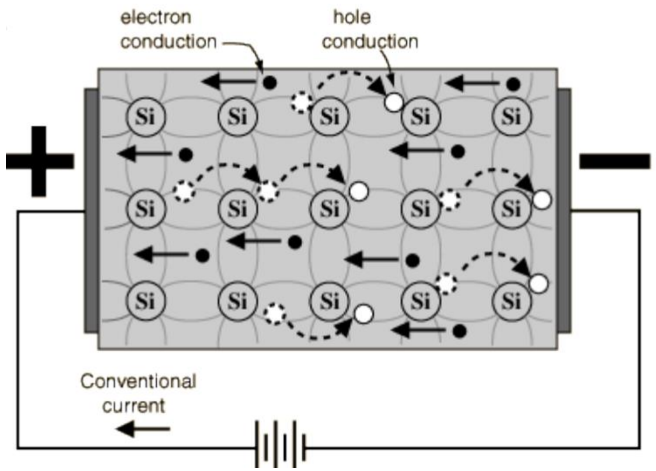
$$\boxed{\tau(T,n)}$$

$$v_d = \left(\frac{dv}{dt}\right)\tau = -\frac{eE}{m^*}\tau$$

$$\boxed{\mu = \frac{v_d}{E} = \frac{e\tau}{m^*}}$$

$$\boxed{\lambda = \langle v\rangle\tau \qquad \text{Mean free path}}$$

$$\lambda(E) = \frac{m^* v_d(E)}{qE}$$



Distance traveled

Mean distance per collision

$$\text{Mean free path estimate} = \frac{\bar{v}t}{\underbrace{\pi d^2}_{\text{Volume of interaction}}\bar{v}t\, n_V} = \frac{1}{\pi d^2 n_V}$$

Number of molecules per unit volume

$$\lambda(E) = \frac{m^* v_d(E)}{qE}$$

v_d saturates at large E and mean free path is 1/
They collide more often with phonons and lose energy
E

Electrons
Equation: $\mathbf{J}_{n,\mathrm{drf}} = \rho_n \mathbf{v}_{dn} = (-en)(-\mu_n \mathbf{E}) = en\mu_n \mathbf{E}$

- $\rho_n = -en$: Electron charge density.
- $\mathbf{v}_{dn} = -\mu_n \mathbf{E}$: Velocity opposite to $\mathbf{E}$.
- Meaning: Current flows parallel to $\mathbf{E}$ despite opposite velocity (due to negative charge).

Holes
Equation: $\mathbf{J}_{p,\mathrm{drf}} = \rho_p \mathbf{v}_{dp} = (ep)(\mu_p \mathbf{E}) = ep\mu_p \mathbf{E}$

- $\rho_p = ep$: Hole charge density.
- $\mathbf{v}_{dp} = \mu_p \mathbf{E}$: Velocity parallel to $\mathbf{E}$.
- Meaning: Current flows parallel to $\mathbf{E}$.

Total drift current;

$$\boxed{J_{drf} = e(\mu_n n + \mu_p p)E}$$

The sum of the individual electron and hole drift current densities

- If **only electrons** are present (n-type semiconductor): current is dominated by J_n.
- If **only holes** are present (p-type semiconductor): current is dominated by J_p.
- In **intrinsic material** (equal numbers of electrons and holes): both contribute.

Total Drift Current Density
Equation: $\mathbf{J}_{\text{drf}} = \mathbf{J}_{n,\text{drf}} + \mathbf{J}_{p,\text{drf}} = e(n\mu_n + p\mu_p)\mathbf{E}$ Meaning: Sum of electron and hole contributions to drift current.

Microscopic Form

$$\boxed{J = \sigma E}$$

- σ: conductivity of the material.
- J: current density ($\mathrm{A\,m^{-2}}$).
- E: electric field ($\mathrm{V\,m^{-1}}$).

Macroscopic Form

$$\boxed{V = IR}$$

- $R = \dfrac{L}{\sigma A}$ is the resistance.
- I: total current, V: potential difference.
- Linear relation between I and V in ohmic regime.

$$\lambda = \langle v \rangle \tau$$

Ohm's law is valid when mean free path, or relaxation time are constant

High E → more scattering → **mobility drops** → **non-Ohmic behaviour**.

Ohm's Law

$\mathbf{J}_{\mathrm{drf}} = \sigma \mathbf{E}$, where $\sigma = e(n\mu_n + p\mu_p)$ is the conductivity.

- **Extrinsic Semiconductors**:
  - n-type ($n \approx N_d \gg p$): $\sigma_n \approx eN_d\mu_n$.
  - p-type ($p \approx N_a \gg n$): $\sigma_p \approx eN_a\mu_p$.
- **Resistivity**: $\rho = 1/\sigma$ (inverse of conductivity, in $\Omega\,\mathrm{cm}$).

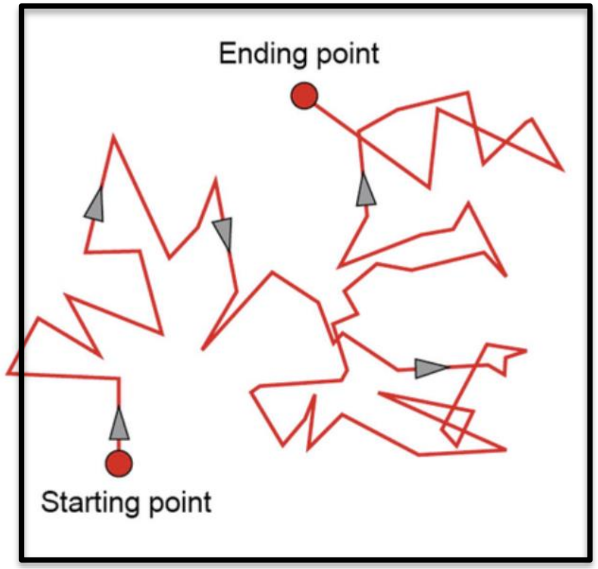
Assumptions for $J = \sigma E$

- Linear relation between J and E (low-field regime).
- Carrier concentration n independent of E (no field ionisation).
- Scattering time τ constant (independent of E and v_d).
- Temperature uniform (no Joule heating).

Diffusion Microscopic Basis

•In gases/liquids

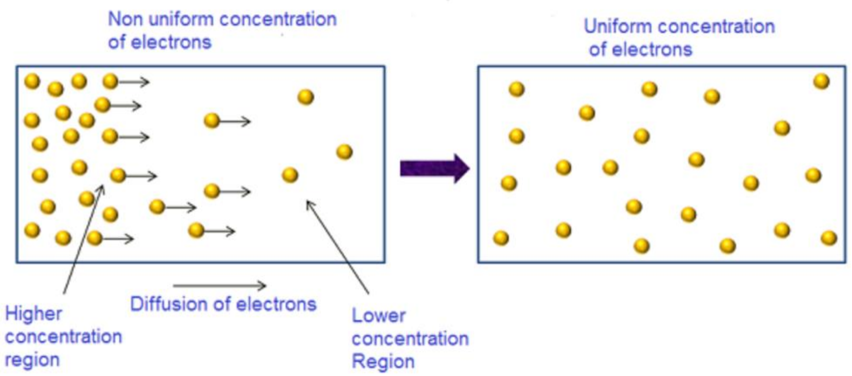
Brownian motion



Temperature effect

Distance quantified by the Einstein Relation

$$\langle x^2 \rangle = 2nDt,$$

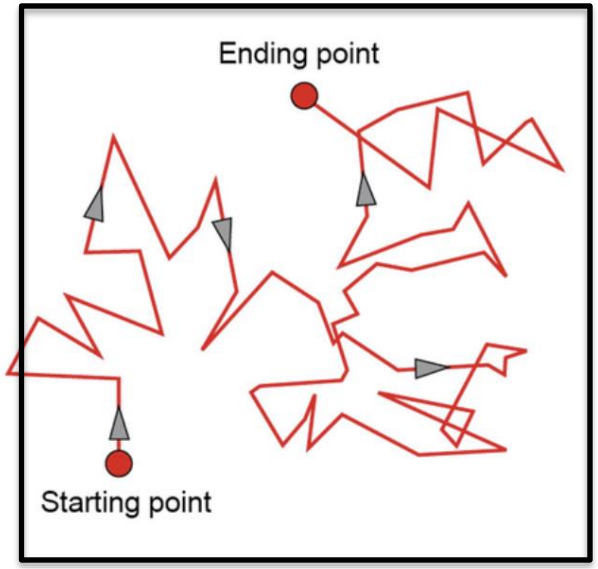


where n is the dimensionality, D is the diffusion coefficient, and t is time.

$$D = l^2/(2n\tau)$$

- l: mean free path (average distance between collisions)
- τ: mean free time between collisions
- n: dimensionality of the system

Diffusion Microscopic Basis



Temperature effect

Entropy (S): A measure of the number of available microstates or, more simply, **disorder**. The second law of thermodynamics states that isolated systems evolve towards **maximum entropy**.

Carrier Distribution and Microstates: A localized, high-concentration carrier distribution has low spatial entropy (highly ordered state).

The Spontaneous Drive: Diffusion is the spontaneous process where a non-uniform distribution expands to fill available space, moving from **low entropy** (ordered) to **high entropy** (disordered) states.

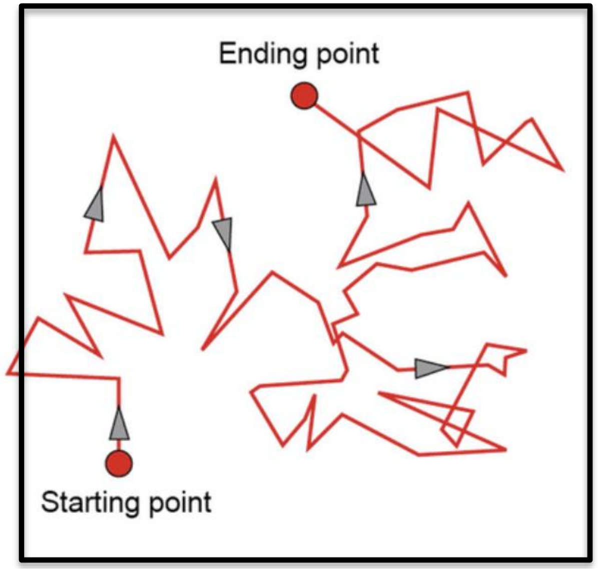
Entropy vs. Chemical Potential: The force driving diffusion is more accurately described by the **gradient of the electrochemical potential**, which includes the chemical potential (μ) related to entropy and the electrostatic potential ($q\psi$).

Non-Equilibrium Drive: In non-equilibrium (e.g., carrier injection), the concentration gradient $\frac{dn}{dx}$ drives the system towards maximum entropy (uniformity) via diffusion.

Point Defect Diffusion

•**Vacancy diffusion:** D limited by **activation barrier** — exponential T-dependence.

$$\langle x^2 \rangle = 2nDt,$$



Temperature effect

$$D = D_0 \exp\left(-\frac{Q}{k_B T}\right) \qquad \frac{1}{\tau} \propto \exp\left(-\frac{Q}{k_B T}\right)$$

Q is the sum of vacancy formation and migration energies

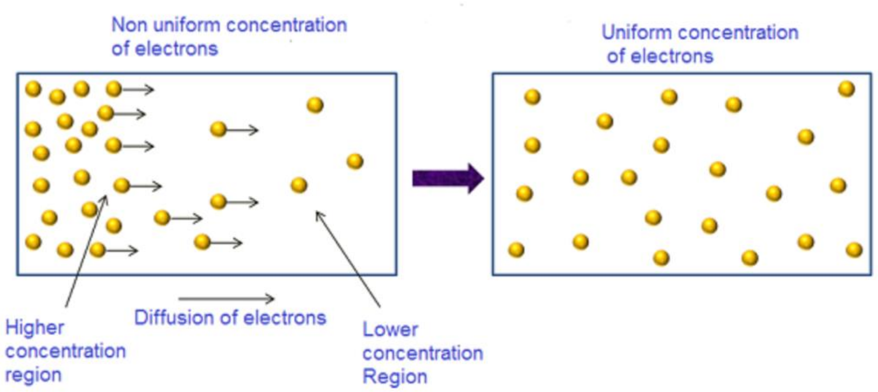
- **Physical Picture:** Atoms hop from one lattice site to another when a neighboring site (vacancy) is available.
- **Key Requirement:** The atom must overcome an energy barrier (migration energy Q_m) to jump, and vacancies must exist (formation energy Q_f).
- **Resulting** D:

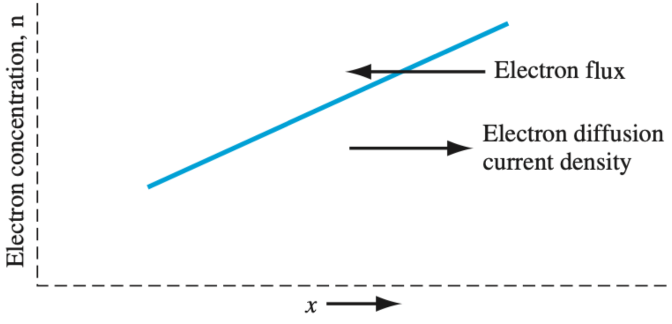
$$D = D_0 \exp\left(-\frac{Q}{k_B T}\right), \quad Q = Q_f + Q_m$$

where D_0 is a pre-factor that depends on attempt frequency and jump distance.

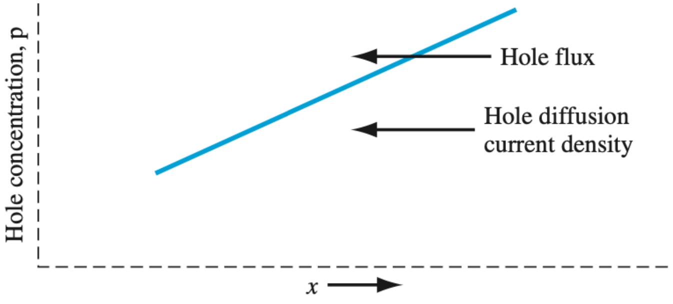
- **Reason:** The probability of a jump follows Boltzmann statistics — higher T means more atoms have enough thermal energy to hop → exponential growth of D.

Diffusion Current First Flick’s Law





(a)



(b)

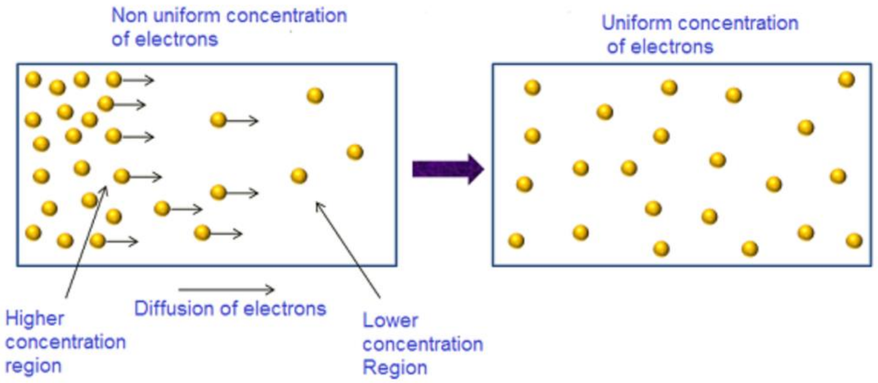
Steady state

$$J_{\text{diff}} = qD\frac{dn}{dx}$$

where J is diffusion flux, n concentration, x position. Negative sign indicates flow from high to low n. Units: D in cm^2/s.2.2 Ficks Second Law (Non-Steady State)

Current flows in opposite direction to the electron motion

Diffusion Current First Flick's Law



Carriers have **random thermal motion** due to temperature. At thermal equilibrium:

- Carriers move randomly in all directions with average speed: $v_{\text{th}} = \sqrt{\frac{3k_B T}{m^*}}$
- In regions of high concentration: more carriers moving in all directions
- In regions of low concentration: fewer carriers moving in all directions
- **Net Result**: More carriers move from high to low concentration regions

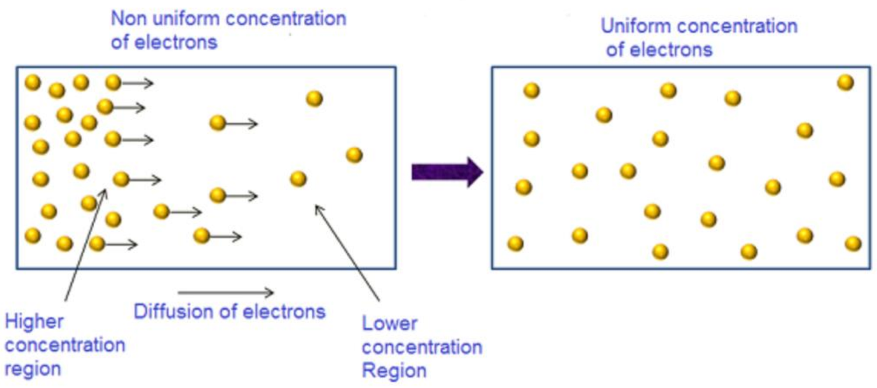
Steady state

$$J_{\text{diff}} = qD\frac{dn}{dx}$$

where J is diffusion flux, n concentration, x position. Negative sign indicates flow from high to low n. Units: D in cm^2/s.2.2 Ficks Second Law (Non-Steady State)

Current flows in opposite direction to the electron motion

Diffusion Current Second Flick's Law



•**Transient (non-steady-state)**

$$\frac{\partial n}{\partial t} = D\frac{\partial^2 n}{\partial x^2}$$

governs $n(x,t)$ for time-varying diffusion. Solution for semi-infinite solid with surface concentration n_s:

$$n(x,t) = n_s - (n_s - n_0)\,\mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

•Lets you estimate how long it takes for a desired concentration level to be reached at a certain depth.

•Fick's second law combined with recombination models predicts how minority carriers diffuse to the junction before recombining.

•**minority carrier diffusion in non-equilibrium conditions**.

Electrons

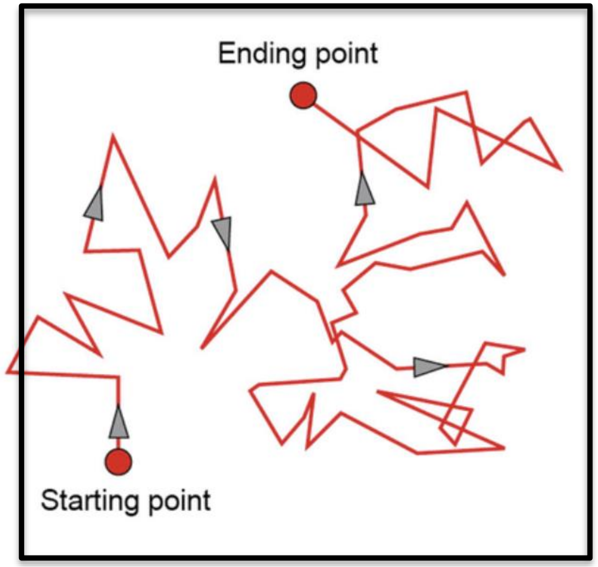
Equation: $\mathbf{J}_n = \mathbf{J}_{n,\text{drf}} + \mathbf{J}_{n,\text{diff}} = en\mu_n\mathbf{E} + eD_n\nabla n$ Meaning: Total electron current as sum of field-driven drift and gradient-driven diffusion.

Holes

Equation: $\mathbf{J}_p = \mathbf{J}_{p,\text{drf}} + \mathbf{J}_{p,\text{diff}} = ep\mu_p\mathbf{E} - eD_p\nabla p$ Meaning: Total hole current as sum of drift and diffusion (note sign for diffusion).

Overall Total Current

$\mathbf{J} = \mathbf{J}_n + \mathbf{J}_p$ Fundamental equations for analyzing semiconductor devices like p-n junctions and transistors.



At thermal equilibrium: No net current flows anywhere in the material.
This requires: $J_{\text{drift}} + J_{\text{diff}} = 0$

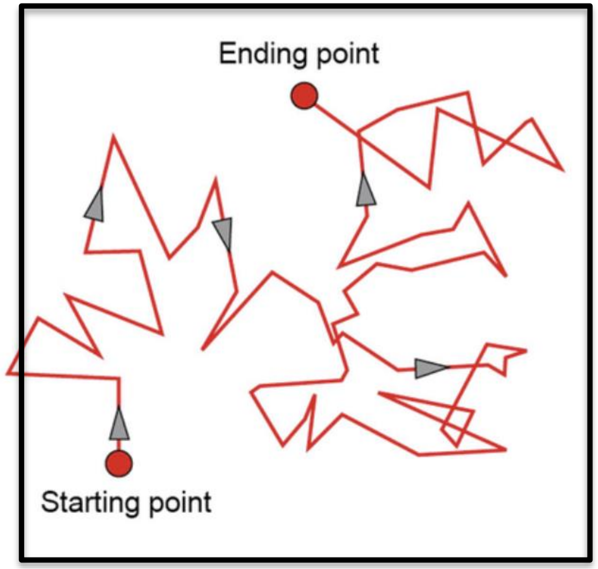
$$q\mu nE - qD\frac{dn}{dx} = 0$$

$$\mu nE = D\frac{dn}{dx}$$

Key Insight: At equilibrium, drift and diffusion perfectly cancel!

Consequence:

- No net current even though both drift and diffusion are acting
- The electric field creates a drift current in one direction
- The concentration gradient creates a diffusion current in opposite direction
- These must exactly balance at equilibrium



At thermal equilibrium, carriers follow Boltzmann statistics:

$$n(x) = n_0 \exp\left(-\frac{qV(x)}{k_B T}\right)$$

where $V(x)$ is the electrostatic potential.

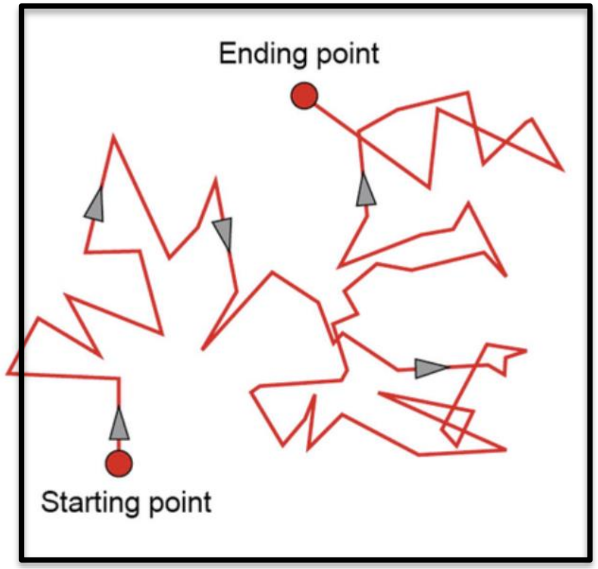
Taking the derivative:

$$\frac{dn}{dx} = n_0 \left(-\frac{q}{k_B T}\right) \exp\left(-\frac{qV(x)}{k_B T}\right) \frac{dV}{dx}$$

$$\frac{dn}{dx} = -\frac{q}{k_B T} n(x) \frac{dV}{dx} = \frac{q}{k_B T} n(x) E$$

where $E = -\frac{dV}{dx}$ is the electric field.

Key Result: The concentration gradient is proportional to the electric field!



Step 1: Substitute Boltzmann into current balance

$$\mu n E = D\frac{dn}{dx} = D \cdot \frac{q}{k_B T} n E$$

Step 2: Cancel n and E (assuming both nonzero)

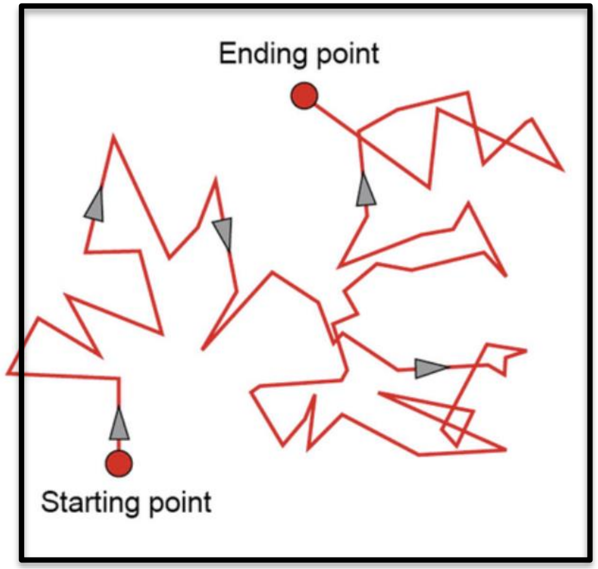
$$\mu = D \cdot \frac{q}{k_B T}$$

Step 3: Rearrange

$$\boxed{D = \frac{k_B T}{q}\mu}$$

Alternative form with thermal voltage $V_T = k_B T/q$:

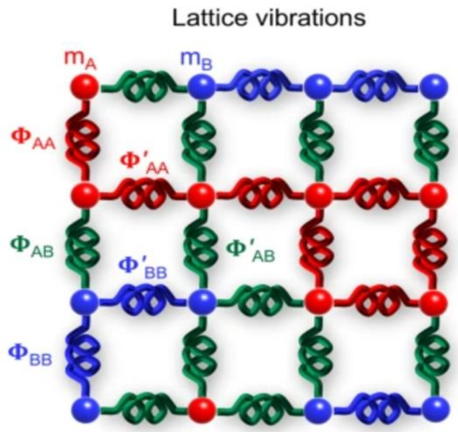
What does $D = \frac{k_B T}{q} \mu$ tell us?

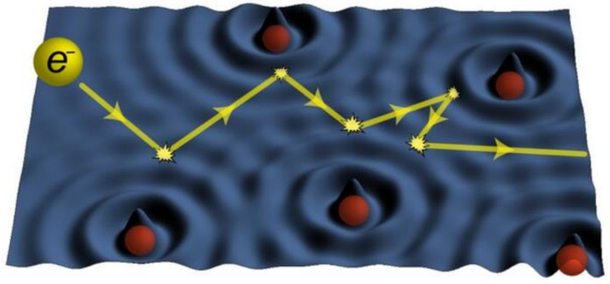


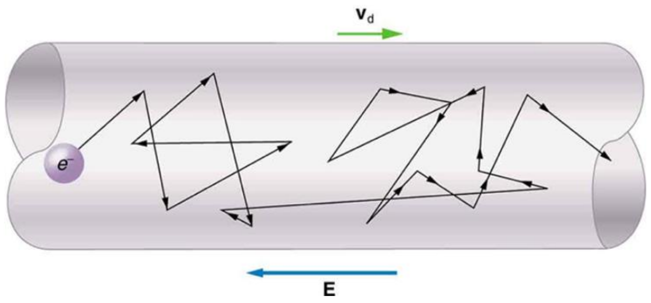
1. **Microscopic Origin:**
   - Drift and diffusion arise from the **same scattering mechanism**
   - Mobility μ measures how easily carriers drift
   - The same μ determines how easily carriers diffuse
   - No separate "diffusion coefficient mechanism"
2. **Thermodynamic Consistency:**
   - Prevents spontaneous currents at equilibrium
   - Ensures $J_{\text{total}} = 0$ when Boltzmann distribution exists
   - If $D/\mu \neq k_B T/q$, the second law of thermodynamics would be violated

Primary Scattering types and effect on mobility

1. **Lattice (Phonon) Scattering** (μ_L):
   - Caused by thermal vibrations of the lattice (phonons).
   - Dominant at higher temperatures ($T > 150$ K).
   - Temperature dependence: $\mu_L \propto T^{-3/2}$.
2. **Ionized Impurity Scattering** (μ_I):
   - Due to Coulomb interactions with ionized donors N_d^+ or acceptors N_a^-.
   - Dominant at lower temperatures.
   - Temperature dependence: $\mu_I \propto T^{3/2}$; inversely proportional to total impurity concentration: $\mu_I \propto N_{\text{total}}^{-1}$.
3. **CarrierCarrier Scattering** (μ_{cc}):
   - Occurs in heavily doped or degenerate semiconductors.
   - Arises from Coulomb interactions among free carriers.
   - Screening reduces scattering efficiency at very high densities.
   - Dependence: $\mu_{cc} \propto n^{-1/3}$; weak temperature dependence near degeneracy.







Primary Scattering types and effect on mobility

Analytical Form

Matthiessens Rule:

$$\frac{1}{\mu(T,n)} = \frac{1}{\mu_{\text{phonon}}(T)} + \frac{1}{\mu_{\text{impurity}}(T,n)} + \frac{1}{\mu_{\text{carrier-carrier}}(n)}$$

Typical Dependencies:

$$\mu_{\text{phonon}} \propto T^{-3/2} \qquad \text{(lattice vibrations)}$$

$$\mu_{\text{impurity}} \propto \frac{T^{3/2}}{n} \qquad \text{(Coulomb scattering)}$$

$$\mu_{\text{carrier-carrier}} \propto n^{-1/3} \qquad \text{(degenerate regime)}$$

Approximate Combined Behaviour:

$$\boxed{\mu(T,n) \approx \left(C_1 T^{3/2} n^{-1} + C_2 T^{-3/2}\right)^{-1}}$$

Primary Scattering types and effect on mobility

Summary

Mobility decreases with both rising temperature and increasing doping.

- At low T: impurity scattering dominates.
- At high T: phonon scattering dominates.
- Maximum μ occurs at intermediate T (transition point).
- Doping increases scattering centres μ falls with n.

High T → more phonons → **mobility decreases**

High n → more impurity and carrier–carrier collisions → **mobility decreases**.

Einstein Relation: $D = \frac{k_B T}{q}\mu(T, n)$

Regime	Main Scattering	$\mu(T, n)$	$D(T, n)$
Low T, High n	Ionised impurity	$T^{3/2}n^{-1}$	$T^{5/2}n^{-1}$
High T, Any n	Lattice (phonon)	$T^{-3/2}$	$T^{-1/2}$
Very High n (deg.)	Carriercarrier	$n^{-1/3}$	$n^{-1/3}T$
Intermediate T	Mixed regime	μ_{max}	D_{max}

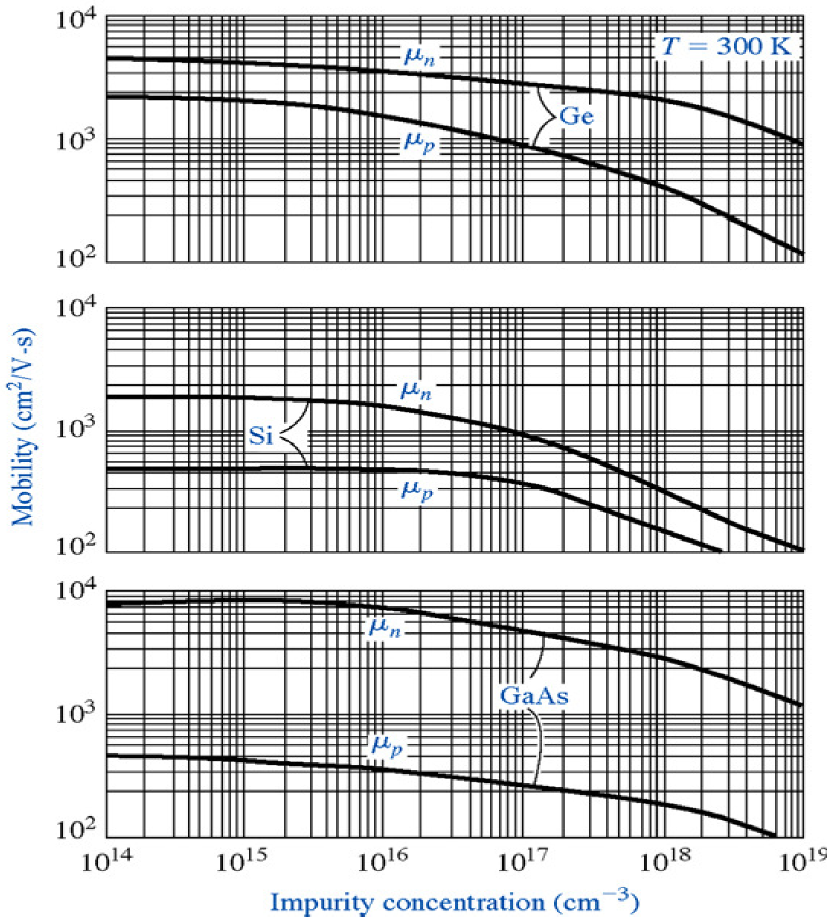
Summary:

- D increases with T at low T (impurities) and decreases at high T (phonons).
- Maximum D occurs where both scattering mechanisms balance.

Ohm's law with T, E and n

Parameter	What increases	What happens microscopically
E	electric field	carriers heat up, scatter more, velocity saturates
T	temperature	more phonons (lattice vibrations) → more scattering
n	doping level	more impurities + carrier–carrier scattering

Anything that makes electrons collide more — higher **field**, higher **temperature**, or higher **doping** — will **reduce mobility**



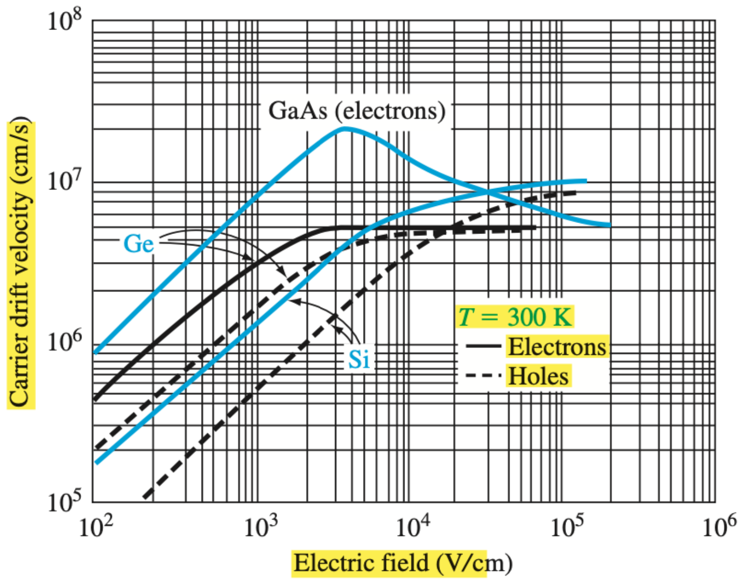
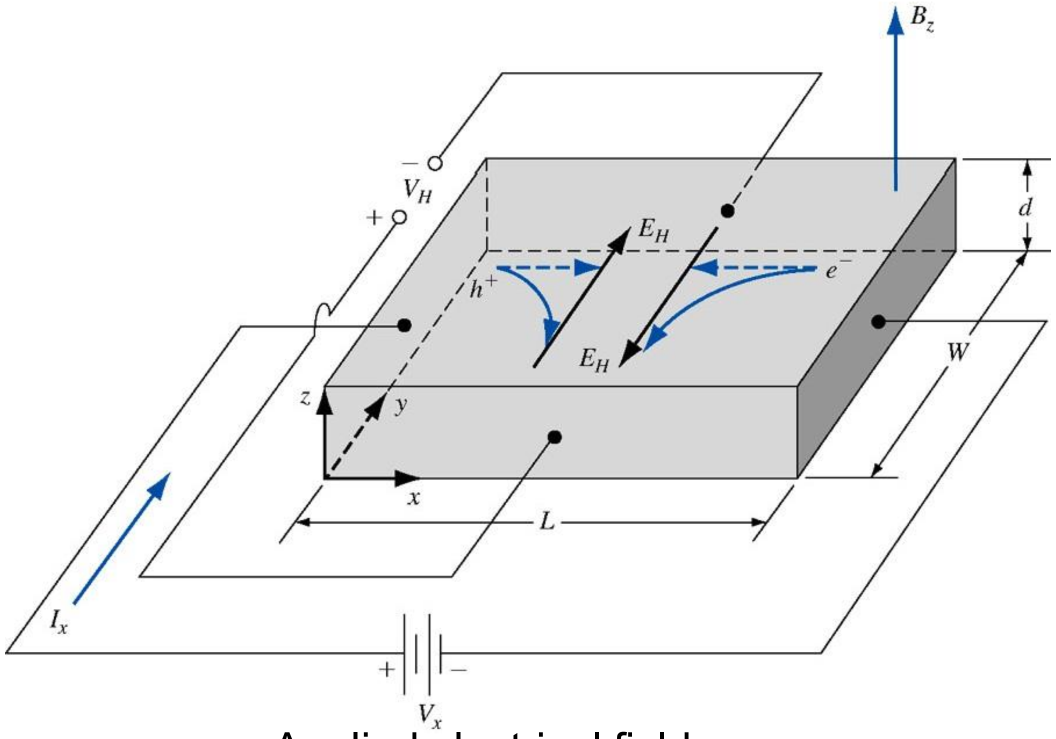


Figure 5.7 | Carrier drift velocity versus electric field for high-purity silicon, germanium, and gallium arsenide. (*From Sze [14].*)

Hall Effect

Hall effect provides the experimental access to diffusivity



Applied electrical field

In the Hall effect, you **apply** a magnetic field and current, **measure** the transverse Hall voltage, and from these you **derive** the carrier type, concentration, and mobility —thereby connecting microscopic transport (diffusion, scattering) with macroscopic measurements.

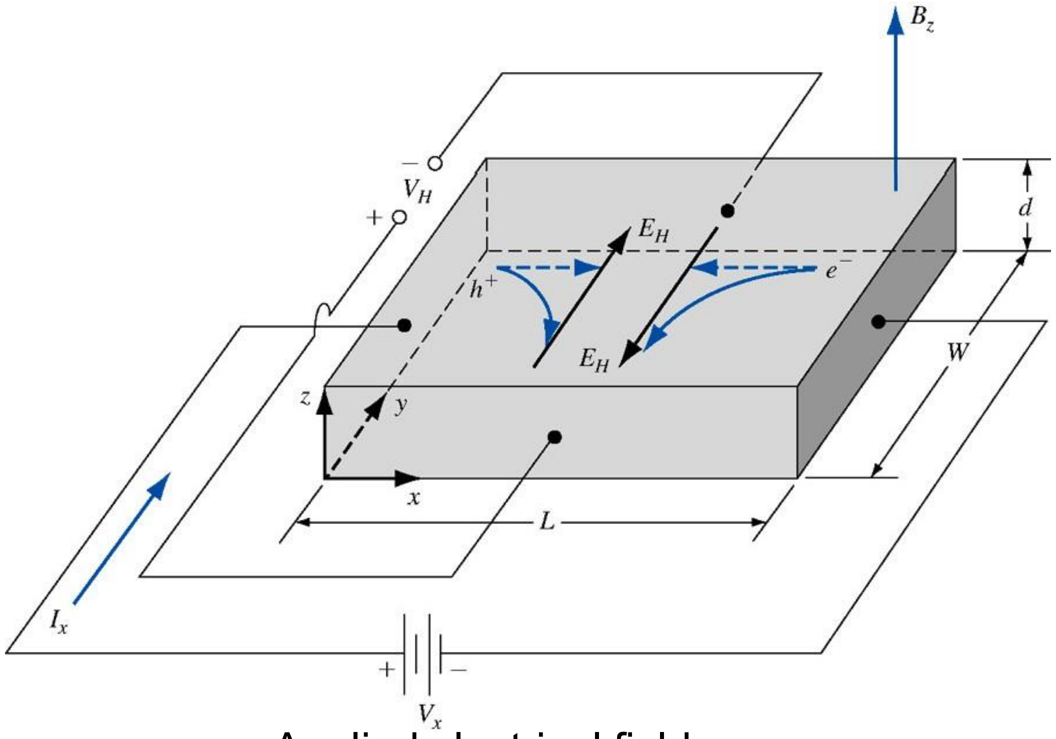
- **Known quantities (controlled experimentally):**
  - Current I
  - Magnetic field B
  - Sample geometry (thickness t, width w, length L)

Forces acting:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$\boxed{E_H = v_x B_z}$$

$$q(E_H+v_xB_z) = 0 \Rightarrow \boxed{E_H = -v_xB_z = -\mu E_xB_z}$$



Applied electrical field

$V_H > 0$ p-type (holes)

$V_H < 0$ n-type (electrons)

1. Force equilibrium:

$$qE_H = qv_xB_z = q\mu E_xB_z$$

2. Hall field and voltage:

$$E_H = \mu E_xB_z \quad \Rightarrow \quad V_H = E_HW = \mu E_xB_zW$$

3. Express drift current:

$$J_x = qn\mu E_x = \frac{I_x}{Wd} \quad \Rightarrow \quad E_x = \frac{I_x}{qn\mu Wd}$$

4. Substitute in V_H:

$$\boxed{V_H = \frac{I_xB_z}{qnd}}$$

(W cancels out)

5. Rearrange:

$$\boxed{n = \frac{I_xB_z}{qdV_H}}$$

$$\boxed{\mu = \frac{\sigma}{qn} = \sigma R_H}$$

where $R_H = V_Hd/(I_xB_z)$.

Example 1: Conductivity in n-type Si

- Given: $N_d = 10^{17}\,\mathrm{cm}^{-3}$, $\mu_n = 1350\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, $e = 1.6 \times 10^{-19}\,\mathrm{C}$.
- Calculation: $\sigma_n \approx eN_d\mu_n = (1.6 \times 10^{-19}\,\mathrm{C}) \cdot (10^{17}\,\mathrm{cm}^{-3}) \cdot (1350\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}) \approx 216\,\mathrm{S\,cm}^{-1}$.
- Resistivity: $\rho_n = 1/\sigma_n \approx 4.63 \times 10^{-3}\,\Omega\,\mathrm{cm}$.

Example 2: Hall Voltage in p-type Si

- Given: $p = 10^{16}\,\mathrm{cm}^{-3}$, $I_x = 1\,\mathrm{mA}$, $B_z = 0.1\,\mathrm{T}$, $W = 0.1\,\mathrm{cm}$, $\mu_p = 480\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$, thickness $t = 0.01\,\mathrm{cm}$.
- $J_x = I_x/(Wt) = 100\,\mathrm{A\,cm}^{-2}$.
- $R_H = 1/(ep) = 6.25 \times 10^3\,\mathrm{cm}^3\,\mathrm{C}^{-1}$.
- $E_y = R_H J_x B_z = 62.5\,\mathrm{V\,cm}^{-1}$.